A manager looks at last month’s sales report and asks three questions in one meeting: “What happened? Why did it happen? What should we do next?” Each question needs a different kind of analysis.

CHAPTER 2 - THE DATA ECOSYSTEM

# Types of Analytics

Lesson 1 of 6

BY THE END

**You can tell descriptive, diagnostic, predictive, and prescriptive questions apart.**

# What is analytics?

**Analytics** means using data to understand what happened, why it happened, what may happen next, or what action to take.

## PLATE I - THE FOUR TYPES OF ANALYTICS

Descriptive

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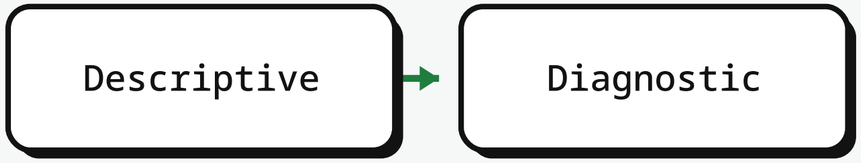

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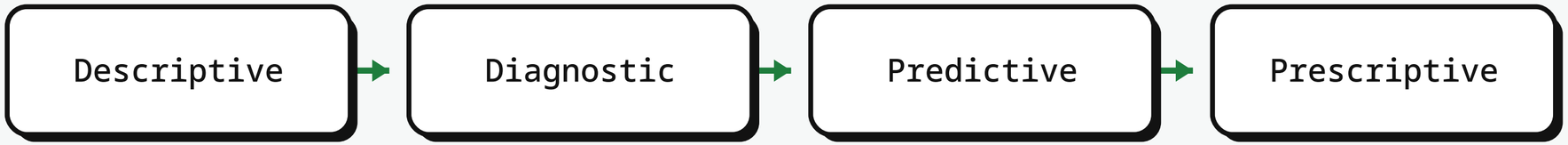


each question needs the one before it answered

# The four types of analytics

| TYPE | QUESTION | EXAMPLE |
|---|---|---|
| Descriptive | What happened? | Rice sales rose 12% last month |
| Diagnostic | Why did it happen? | They rose because the Hargeisa shop stayed open through Ramadan |
| Predictive | What might happen next? | Sales may stay high next month if the longer hours continue |
| Prescriptive | What should we do? | Keep the longer hours in Hargeisa, and try them in Muqdisho |

# Simple way to remember them

- **Descriptive** = report the past
- **Diagnostic** = explain the past
- **Predictive** = forecast the future
- **Prescriptive** = recommend action

Most beginners start with descriptive analytics, such as summaries, totals, and charts. More advanced work adds prediction and recommendation.

# The four are cumulative

You cannot explain why rice sales rose until you know that they rose. You cannot forecast next month without knowing what drove last month. And a recommendation made without the three steps underneath it is just an opinion with a number attached to it.

4 - WHY THIS MATTERS

So when an organization says it wants prediction, the honest first question is whether it can yet describe what already happened.

KEY TERMS

| TERM | DEFINITION |
| --- | --- |
| Analytics | The use of data to find patterns and support decisions |
| Descriptive analytics | Summarizes what already happened |
| Diagnostic analytics | Explains why something happened |
| Predictive analytics | Estimates what may happen in the future |
| Prescriptive analytics | Recommends actions based on analysis |